Supplementary Figure S1

The 84 detected maps of experiment E1 and the paths of TVE, WDT and the visibility graph

Manuscript robotics-4632217, revision 1 (O. Penagos, I. Ladino, C. R. Suárez).

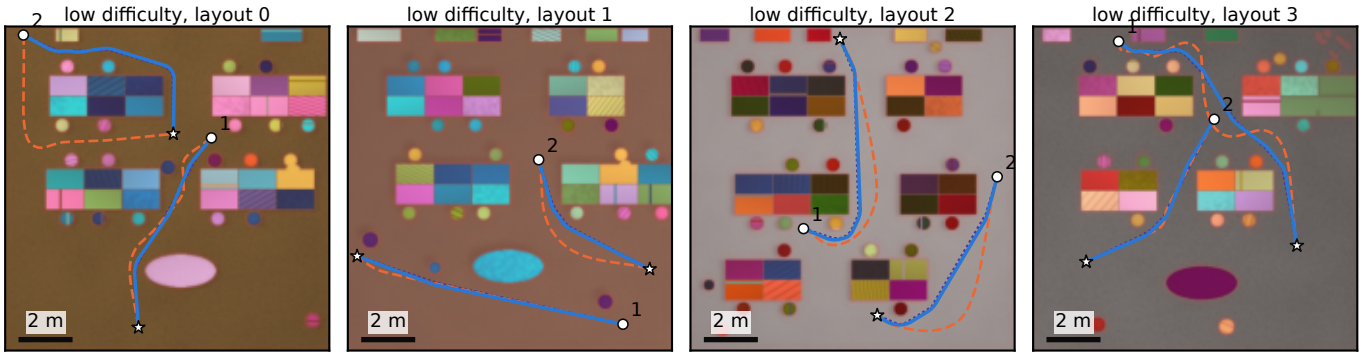
One page per type of scene (office, warehouse, apartment, workshop, retail store, outdoor plaza and hospital ward). Rows: low, medium and high difficulty of the image; columns: layouts 0 to 3.

Each panel shows the rectified camera image (400 x 400 px), the obstacles detected by the image processing (red), the band added by the dilation by the robot radius (light red), and, for the two queries of the map (1 and 2: circle = S, star = T), the paths of TVE (solid blue), WDT (dashed orange) and the visibility graph (dotted purple), whose length is the reference L*. Below each panel: the length ratio L/L* of TVE and WDT for each query. The bar measures 2 m; the floor sides are 12, 16, 10, 12, 14, 18 and 14 m, in the order above.

The planners are executed again with the parameters of E1 (configs/), and the ratios printed in the panels were checked against results/raw/e1.csv: 336 values, largest difference 3.1e-08.

Produced by figures/make_supplementary.py of the reproducibility package.

Figure S1 (Office): detected maps of E1 and paths (L/L* of TVE and WDT for each query)

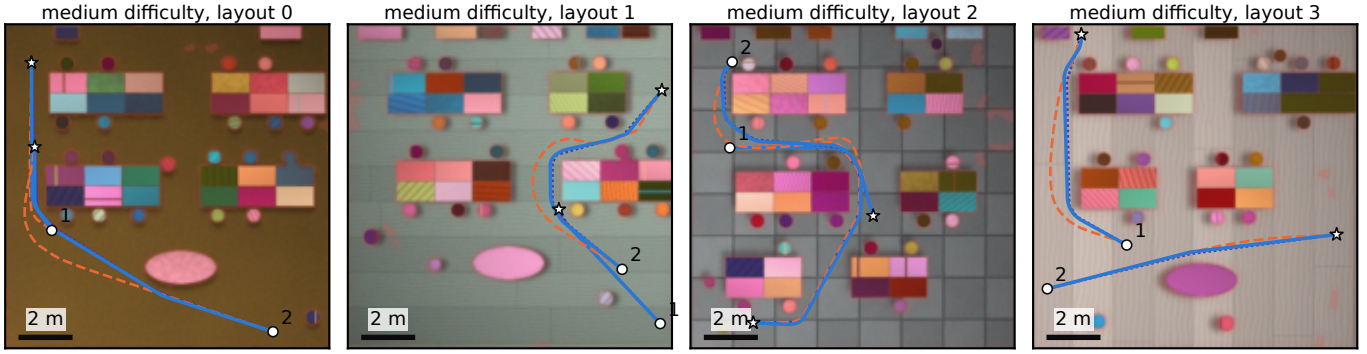


q1: TVE 1.005, WDT 1.058
q2: TVE 1.013, WDT 1.170

q1: TVE 1.001, WDT 1.005
q2: TVE 1.008, WDT 1.097

q1: TVE 1.012, WDT 1.080
q2: TVE 1.011, WDT 1.145

q1: TVE 1.012, WDT 1.180
q2: TVE 1.007, WDT 1.015

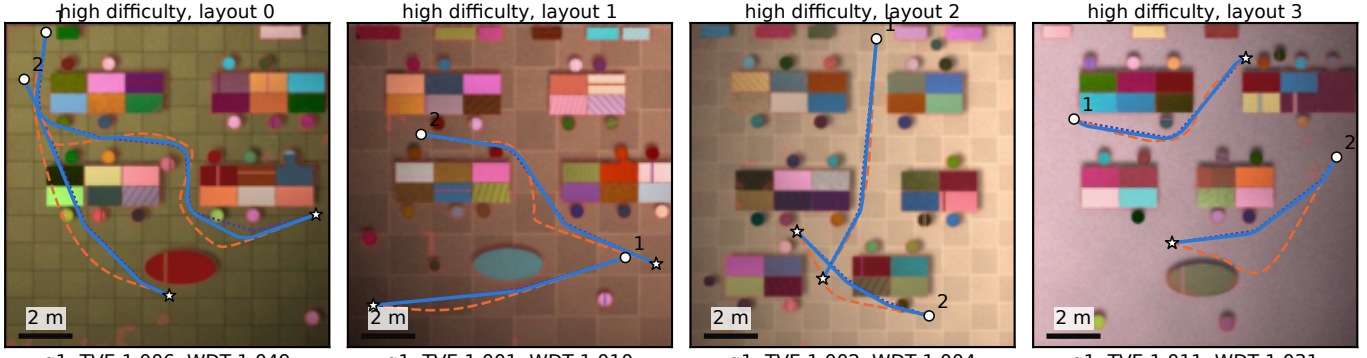


q1: TVE 1.007, WDT 1.027
q2: TVE 1.004, WDT 1.068

q1: TVE 1.005, WDT 1.000
q2: TVE 1.014, WDT 1.134

q1: TVE 1.001, WDT 1.065
q2: TVE 1.012, WDT 1.052

q1: TVE 1.012, WDT 1.067
q2: TVE 1.001, WDT 1.002



q1: TVE 1.006, WDT 1.049
q2: TVE 1.023, WDT 1.132

q1: TVE 1.001, WDT 1.010
q2: TVE 1.006, WDT 1.068

q1: TVE 1.002, WDT 1.004
q2: TVE 1.006, WDT 1.075

q1: TVE 1.011, WDT 1.031
q2: TVE 1.005, WDT 1.198

TVE (proposed) WDT (original method) Visibility graph (optimum L*)

Figure S1 (Warehouse): detected maps of E1 and paths (L/L* of TVE and WDT for each query)

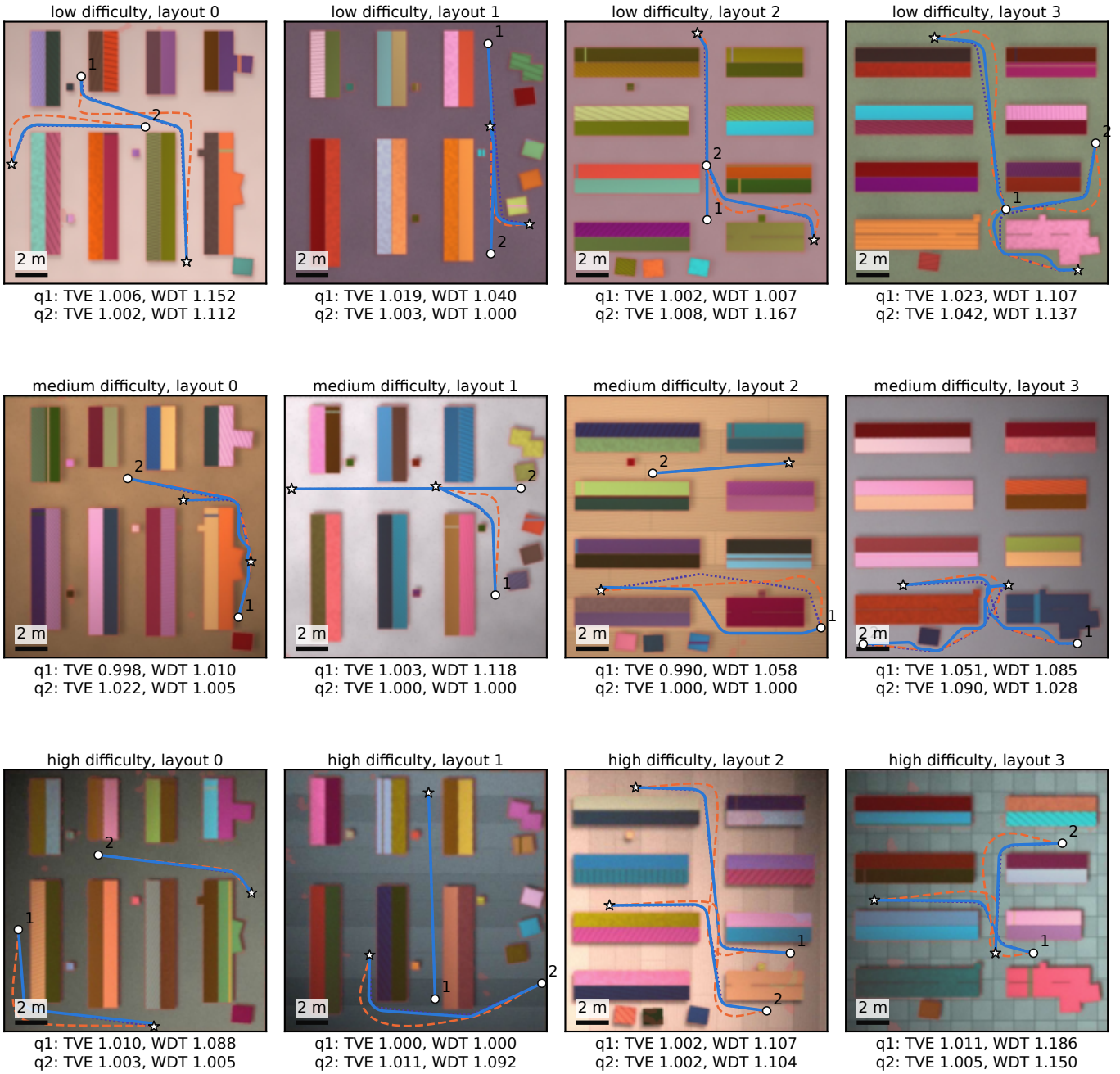


Figure S1 (Apartment): detected maps of E1 and paths (L/L* of TVE and WDT for each query)

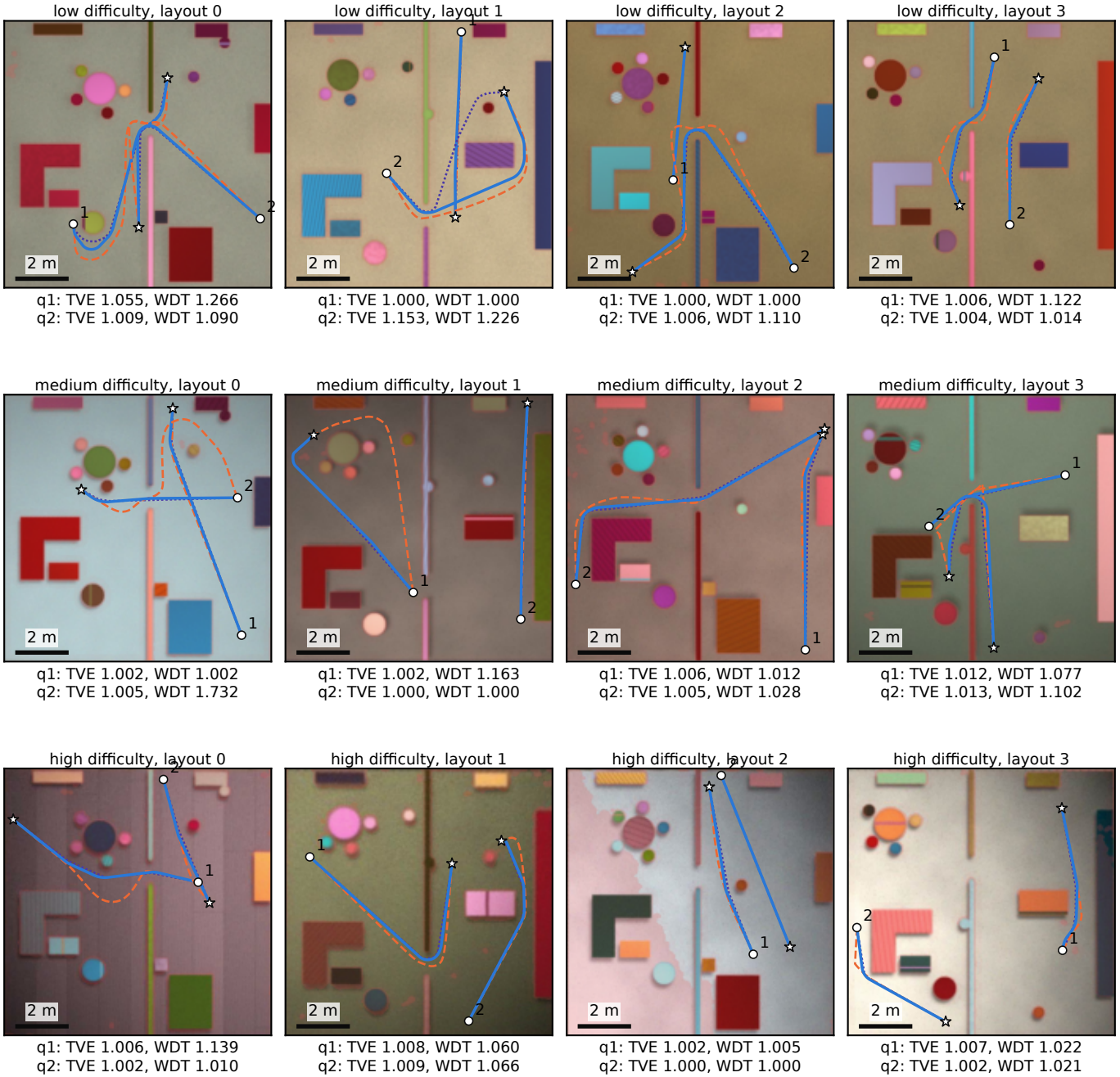


Figure S1 (Workshop): detected maps of E1 and paths (L/L* of TVE and WDT for each query)

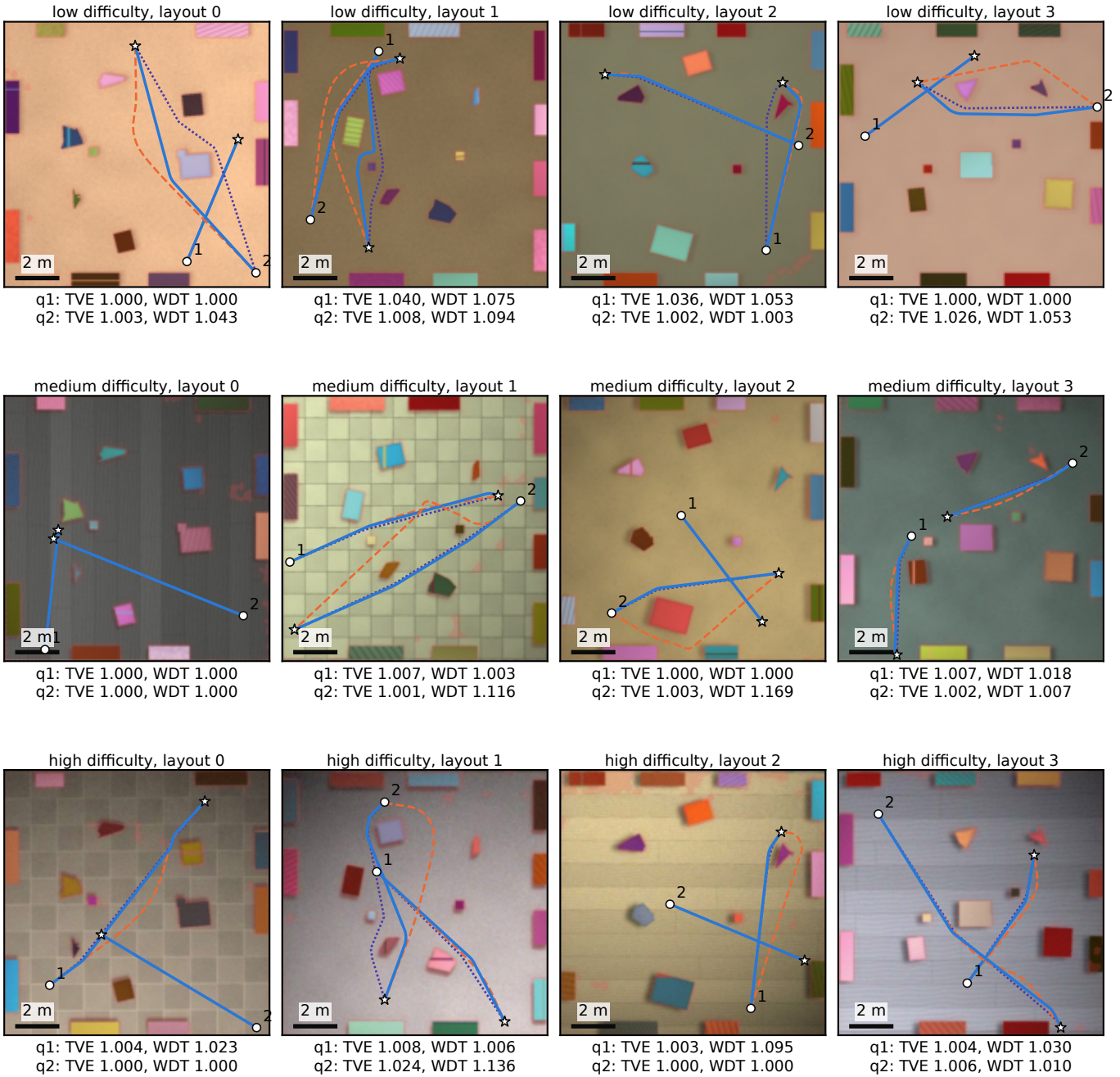
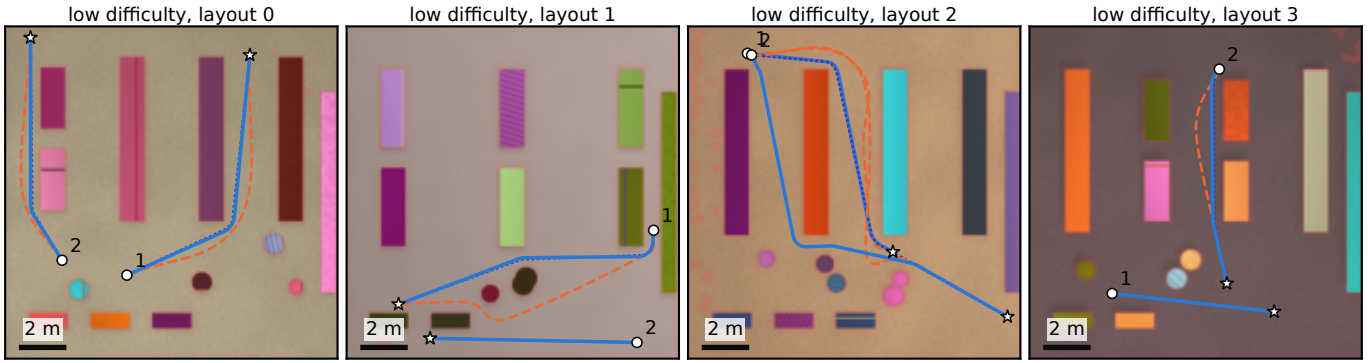


Figure S1 (Retail store): detected maps of E1 and paths (L/L* of TVE and WDT for each query)

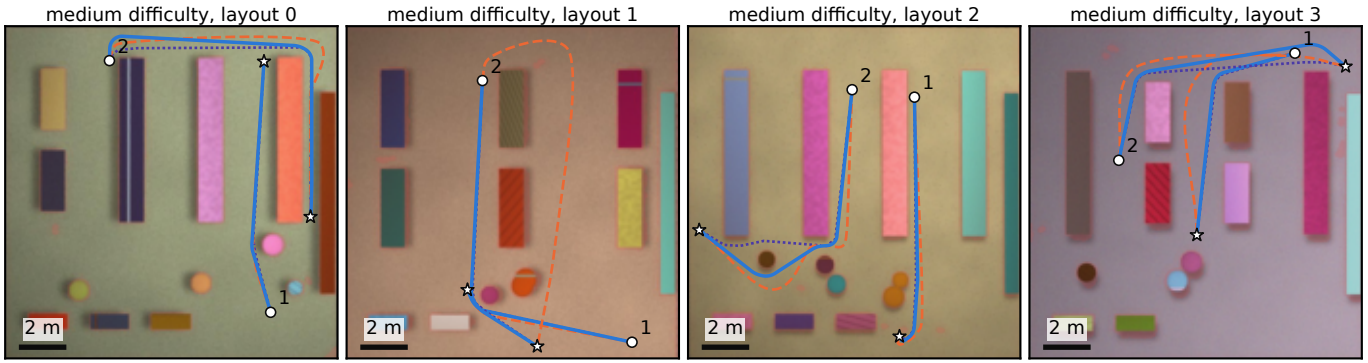


q1: TVE 1.003, WDT 1.033
q2: TVE 1.002, WDT 1.017

q1: TVE 1.002, WDT 1.048
q2: TVE 1.000, WDT 1.000

q1: TVE 1.003, WDT 1.098
q2: TVE 1.012, WDT 1.115

q1: TVE 1.000, WDT 1.000
q2: TVE 1.006, WDT 1.028

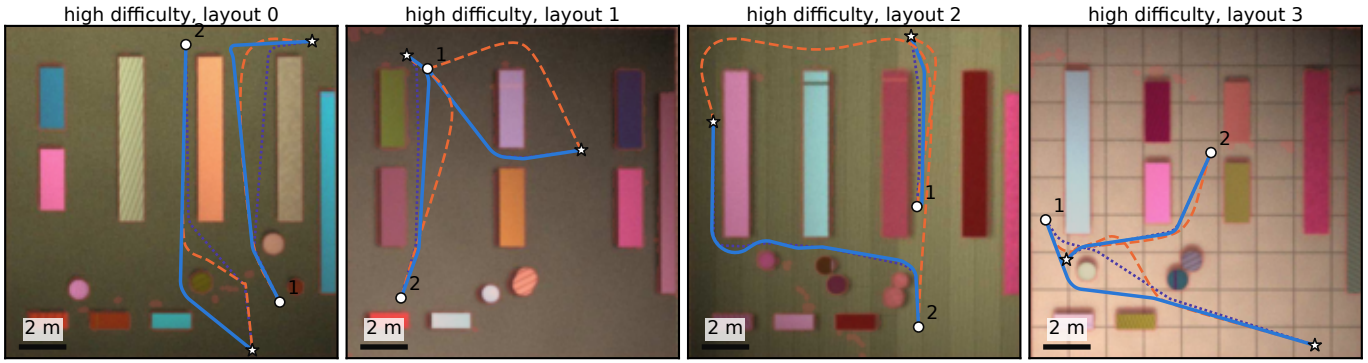


q1: TVE 1.002, WDT 1.002
q2: TVE 1.038, WDT 1.075

q1: TVE 1.004, WDT 1.011
q2: TVE 1.002, WDT 1.368

q1: TVE 0.980, WDT 1.031
q2: TVE 1.068, WDT 1.177

q1: TVE 1.005, WDT 1.108
q2: TVE 1.034, WDT 1.055



q1: TVE 1.142, WDT 1.083
q2: TVE 1.034, WDT 1.041

q1: TVE 1.001, WDT 1.128
q2: TVE 1.030, WDT 1.083

q1: TVE 1.029, WDT 1.074
q2: TVE 1.022, WDT 1.511

q1: TVE 1.062, WDT 1.093
q2: TVE 1.008, WDT 1.034

TVE (proposed) WDT (original method) Visibility graph (optimum L*)

Figure S1 (Outdoor plaza): detected maps of E1 and paths (L/L* of TVE and WDT for each query)

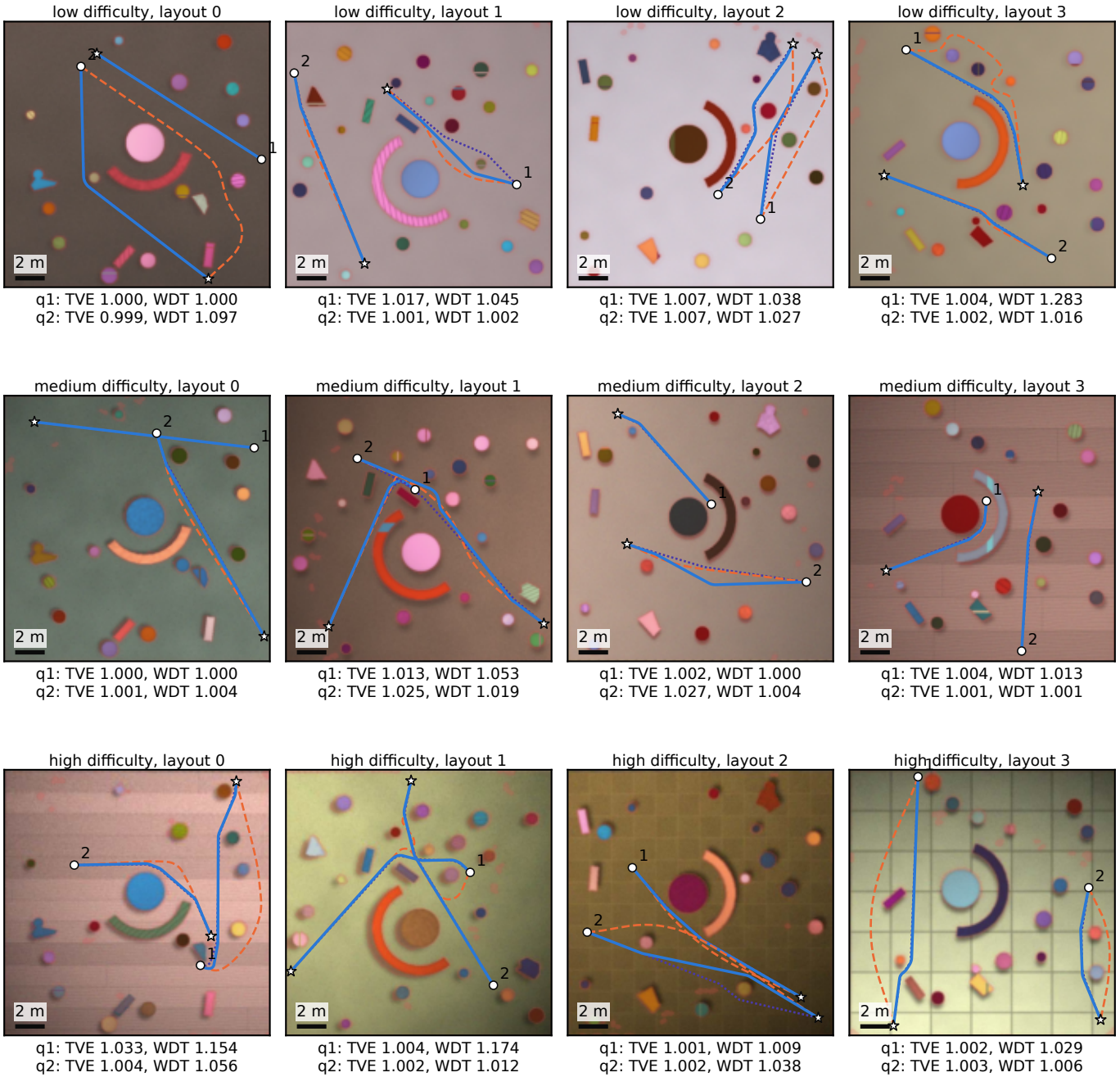


Figure S1 (Hospital ward): detected maps of E1 and paths (L/L* of TVE and WDT for each query)

